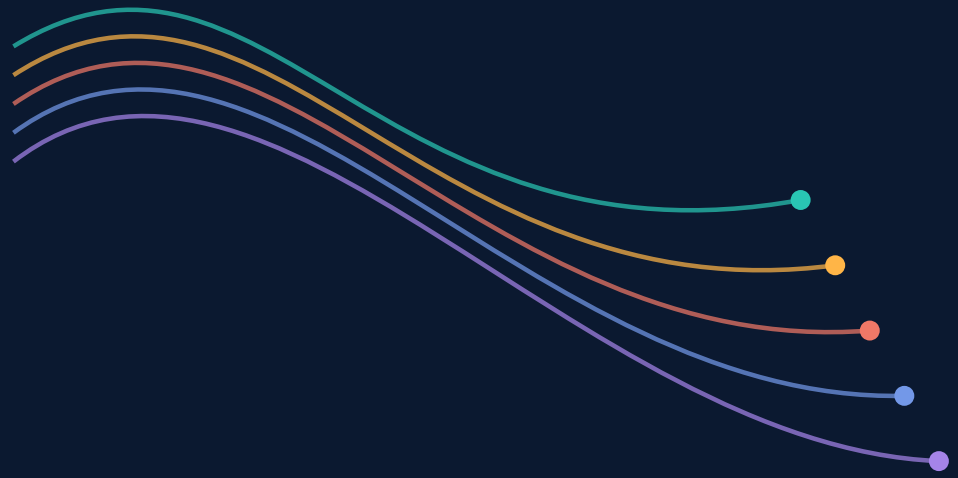
TMS / DATA & ML

Know before your clients do.



A flight early-warning system for a New York travel agency, built on 435,352 departures from JFK, LGA and EWR. [01]

2 hours
notice before scheduled departure
[05]

2 alerts
late-or-cancelled / cancellation
[06]

1 year
real flight history · 2023
[01]

THE PROPOSAL

Pilot two human-reviewed alerts, then decide with evidence—not assumptions.

THE BUSINESS CASE

A bad flight is a client problem first

Give the agency time to warn, check connections and prepare alternatives.

25.7%

TRAINING-PERIOD RISK

Flights late or cancelled, Jan–Sep; 330,883 scheduled flights.

[01]

10,738

CANCELLATIONS

Every cancellation can force a rebooking decision.

[01]

3 airports

NEW YORK DEPARTURES

JFK · LaGuardia (LGA) · Newark (EWR).

[01]

THE TWO QUESTIONS

Will this flight be late or cancelled?

Should the agency prepare a rebooking?

Both scores are intended for the two-hour point. [05]

A person verifies live airline status before acting.

WHAT THE DATA REVEALED

Risk builds, clusters—and is not just weather

Three findings with a business use. Training-period analysis unless stated. [02, 06]

THE DAY GETS HARDER

13.4% [02]
scheduled 05–08 h



39.9% [02]
scheduled 19–22 h



+26.5 percentage points. [02]

CANCELLATIONS CLUSTER ON A FEW DAYS

27 / 273 days [02]

accounted for 57.5% of cancellations. [02]

THE BEST WARNING IS WHAT JUST HAPPENED AT THE AIRPORT

recent airport-state importance
0.228

weather
0.023

About 10× stronger in this validation importance comparison. [06]

TARGET THE RESPONSE

Destinations have different reliability profiles

Three clustered groups plus a small-airport label; use the groups as guidance, not certainty.

SHORT-HAUL

57 airports

101,287 flights

21.2% late

Prepare rebooking options; cancellation rate was 4.3%.

[03]

MAIN LONG-HAUL

45 airports

226,896 flights

27.8% late

Prioritize afternoon and evening connection checks.

[03]

CHRONIC DELAYS

7 airports

2,374 flights

53.1% late

Warn early and check onward connections.

[03]

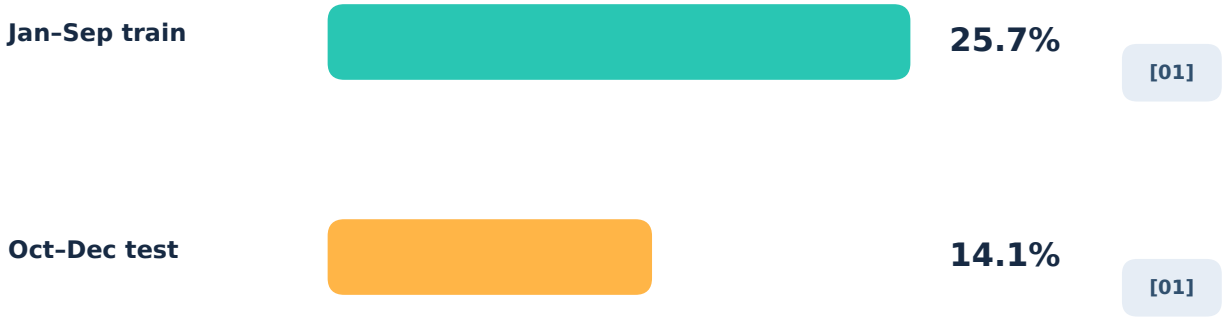
CAUTION The structure is modest (silhouette 0.322). Avoid strong claims about small groups. [03]

MEASURE WHAT MATTERS

The year changes the operating picture

Use a small set of explicit, denominator-aware reliability measures.

LATE OR CANCELLED



Cancellation rate: 3.18% train → 0.21% test. [01]
This drift is why test performance must be shown against its own prevalence.

THE FIVE METRICS

M1	on-time rate	departed flights
M2	mean departure delay	minutes · departed
M3	cancellation rate	all scheduled flights
M4	>60 min late	departed flights
M5	late or cancelled	all scheduled flights

Full-year checks: M1 79.03% · M2 13.84 min · M3 2.47% · M4 8.41% · M5 22.91%. [04]

BUILT TO BE TRUSTED

Score only with information available at decision time

The prediction point is two hours before scheduled departure. [05]



SCHEDULE & ROUTE

Calendar · carrier · origin/destination
Scheduled flight volume

RECENT AIRPORT STATE

Only earlier departures known by T
No earlier cancellation flag

WEATHER & AIRCRAFT

Strict weather is available by T
Scheduled-hour weather is a forecast stand-in

Leakage guardrails: exclude post-departure outcomes and tail-missing flags; fill unknown plane data using Jan–Sep carrier statistics. [05]

AN HONEST TEST

Tune on validation; keep the test months unseen

Time order mirrors how the system would be built and checked in practice.

JAN–JUN

Fit candidates
222,402 flights

[06]

JUL–SEP

Tune and choose
108,481 flights

[06]

OCT–DEC

Final test · once
104,469 flights

[06]

Then refit selected settings on Jan–Sep: 330,883 flights. [06]

WHY NOT ACCURACY?

“Nobody is late” is 85.9% accurate on test and finds no late flights. [06]

We compare PR-AUC with the dumb answer on the same period; prevalence is the baseline. [06]

THE RESULT

Both alerts rank risk above the dumb baseline

Precision–recall AUC on held-out Oct–Dec flights; intervals reflect day-level resampling. [06]

PREDICTION	DUMB BASELINE	MODEL PR-AUC	95% INTERVAL	READOUT
Late or cancelled [06]	**0.141**	**0.339**	0.301–0.376	2.4× prevalence
Cancelled [06]	**0.0021**	**0.120**	0.002–0.233	rare-event caution

Cancellation scores are useful for prioritization—not certainty. The interval is wide because the test set had only 224 cancellations. [01, 06]

REFRESH REQUIRED / Stored model outputs predate the D10 imputation update; rerun notebooks 05–07 before final submission. [05–07]

DECISION CARD / ALERT 1

Warn clients about a possible delay

Late-or-cancelled score · test quarter · human review before messaging. [07]

.597
validation cutoff
[07]

11.5%
of test flights flagged
[07]

37.4%
of flags were right
[07]

30.5%
of problem flights caught
[07]

WHAT THE TEST QUARTER LOOKED LIKE

11,974 flags

4,481 true positives

About 130 flagged flights per day; lift 2.7× over a random shortlist. [07]

ACTION + HUMAN CHECK

Message · check connections

Verify live airline status first. Mornings are a weak spot: 5.8% recall. [07]

REFRESH REQUIRED / Model outputs predate the D10 update; rerun notebooks 05–07 before final submission. [05–07]

DECISION CARD / ALERT 2

Prepare rebooking for cancellation risk

A prioritization list, not a promise that a flight will be cancelled. [07]

.137
validation cutoff
[07]

484
test flags
[07]

19.4%
flag precision
[07]

42%
cancellations caught
[07]

94 / 224 cancellations identified

5.3 flags per day · lift 91× over random. [07]

Offer rebooking options; never auto-rebook.

HANDLE WITH CARE

115 of the 224 test cancellations fell on one day.

Only one test quarter and 224 events: the measured result is highly uncertain. [01, 07]

REFRESH REQUIRED / Model outputs predate the D10 update; rerun notebooks 05–07 before final submission. [05–07]

LIMITS & MONITORING

Keep a person in the loop—and watch drift

The test quarter is not a guarantee of future performance.

SEASONAL DRIFT

Late-or-cancelled rate fell from 25.7% in train to 14.1% in test. [01]

EARLY-MORNING GAP

Alert 1 caught 5.8% of morning problem flights; Newark cancellation recall was 0%. [06, 07]

RARE EVENT

115 / 224 test cancellations landed on 18 December; cause is unexplained. [01, 07]

A TENTATIVE CHECK RULE

Alert 1

Review if weekly recall is below 14% for two weeks running. [07]

Alert 2

Review 4-week blocks with ≥30 cancellations if recall is below 23%. [07]

Thresholds are tentative; the evidence is limited.

Model score snapshot predates D10 correction; refresh downstream analyses before submission. [05–07]

THE DECISION

Approve a careful pilot. Earn the right to scale.

Start with human-reviewed alerts. Measure capacity, outcomes and drift before changing client operations.

01 REVIEW — A named person verifies live status before any client contact.

02 MONITOR — Check performance weekly; pause and investigate when the stop rule triggers.

03 IMPROVE — Add real forecast data and cancellation timestamps when available.

THE ASK

Agree the daily alert capacity and who owns the review.

Pilot duration is a business choice—not a model result.

Data: nycflights23 (CC0). Results, limitations and methods are traceable in notebooks 01–07. [01–07]